

TASK 2: Chatbot for FAQs using Machine Learning

Git Repository Link: https://github.com/rooparooa1029-web/CodeAlpha_FAQChatbot/new/main

Aim

To develop a chatbot that answers frequently asked questions using Natural Language Processing (NLP) and Machine Learning techniques.

Introduction

FAQ Chatbot is a Natural Language Processing (NLP) project that automatically answers user questions based on predefined FAQs.

The chatbot:

- Accepts user questions
- Processes text using NLP
- Finds the most similar FAQ question
- Displays the best matching answer

This project helps in understanding:

- NLP
- Text preprocessing
- Cosine similarity
- Chatbot development
- Intent matching

Tools and Libraries Used

- Python
- Pandas
- NLTK
- Scikit-learn
- NumPy

Official library websites:

- [NLTK](#)
- [Scikit-learn](#)

Dataset Description

The dataset contains:

- Questions

- Answers

Example FAQ dataset:

Question	Answer
What is AI?	AI stands for Artificial Intelligence
What is Python?	Python is a programming language
What is Machine Learning?	ML enables computers to learn from data

Dataset File

faq_data.csv

Question,Answer

What is AI?,AI stands for Artificial Intelligence.

What is Python?,Python is a programming language.

What is Machine Learning?,Machine Learning enables computers to learn from data.

What is NLP?,NLP stands for Natural Language Processing.

What is Chatbot?A, chatbot is a software application that interacts with users.

Machine Learning Steps

1. Import libraries
2. Load FAQ dataset
3. Preprocess text using NLP
4. Convert text into vectors
5. Calculate cosine similarity
6. Match user question with FAQ
7. Display best matching answer

Python Program

```
import pandas as pd
```

```
import nltk
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
from sklearn.metrics.pairwise import cosine_similarity
```

```
# Download tokenizer
```

```
nltk.download('punkt')
```

```
# Load dataset
```

```
data = pd.read_csv("faq_data.csv")
```

```
# Questions and Answers
```

```
questions = data['Question']
```

```

answers = data['Answer']

# Convert text into vectors
vectorizer = TfidfVectorizer()

question_vectors = vectorizer.fit_transform(questions)

# Chatbot function
def chatbot():

    print("FAQ Chatbot")
    print("Type 'exit' to stop\n")

    while True:

        user_question = input("You: ")

        if user_question.lower() == "exit":
            print("Chatbot: Goodbye!")
            break

        # Convert user question into vector
        user_vector = vectorizer.transform([user_question])

        # Calculate similarity
        similarity = cosine_similarity(
            user_vector,
            question_vectors
        )

        # Best match
        index = similarity.argmax()

        # Display answer
        print("Chatbot:", answers[index])

# Run chatbot
chatbot()

```

Output

```

FAQ Chatbot
Type 'exit' to stop

```

```

You: What is AI?
Chatbot: AI stands for Artificial Intelligence.

```

```

You: What is NLP?
Chatbot: NLP stands for Natural Language Processing.

```

You: exit

Chatbot: Goodbye!

Explanation of the Code

1. Import Libraries

Libraries for NLP and Machine Learning are imported.

2. Load Dataset

FAQ dataset is loaded using Pandas.

3. Text Vectorization

TF-IDF converts text into numerical vectors.

4. Cosine Similarity

Similarity between user question and FAQ questions is calculated.

5. Best Match Selection

The chatbot selects the most similar question.

6. Display Answer

The corresponding answer is displayed to the user.

Advantages

- Beginner friendly
- Fast response generation
- Uses NLP techniques
- Easy to customize
- Can handle multiple FAQs

Applications

- Customer support
- Educational assistants
- E-commerce websites
- Banking support systems
- Help desk automation

Conclusion

The FAQ Chatbot successfully answers user questions using NLP and cosine similarity techniques. The project demonstrates how Machine Learning and text processing can be used to build intelligent chatbot systems.

